Lecture 19

Uniform Plane Waves in Free Space

Dr. Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard, Brian Cunningham, and Victoria Shao
- Copying, publishing or distributing without express written permission is prohibited.

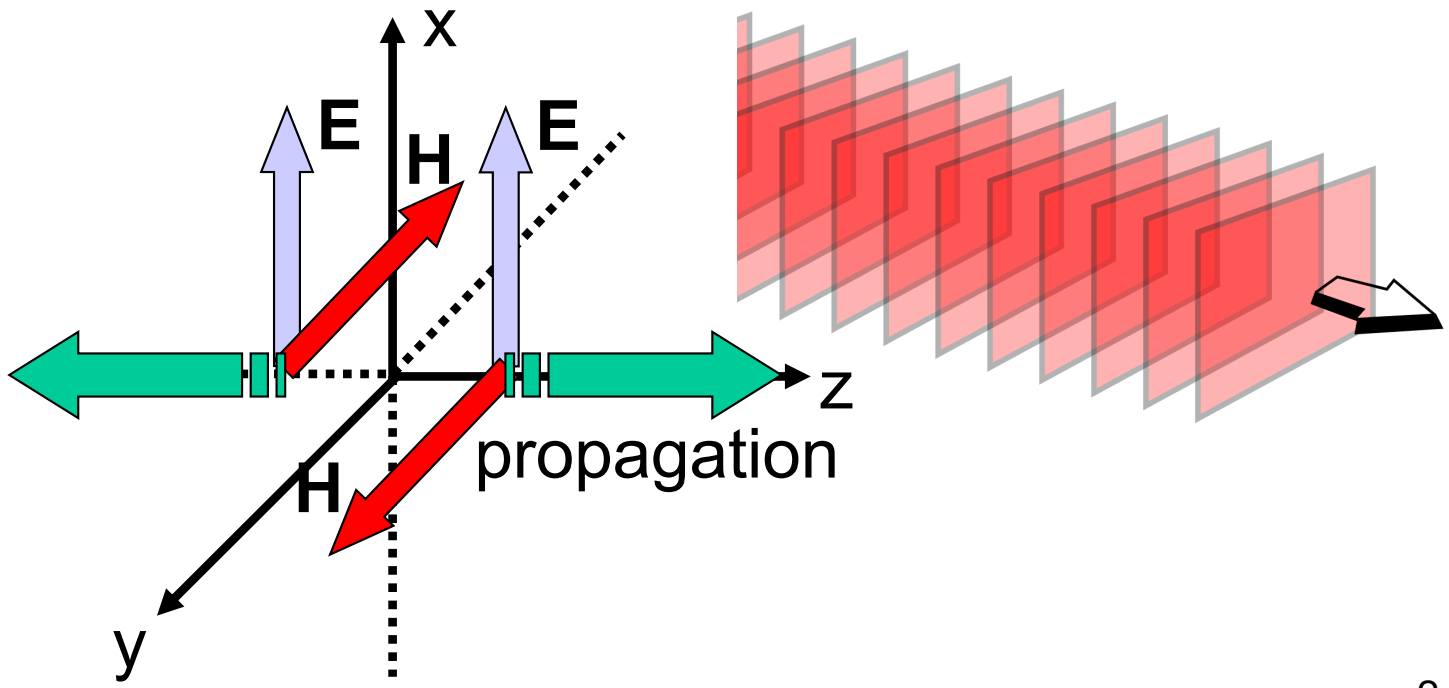
Summary so far …

- We have **E** and **H** that are
  - Perpendicular to each other
  - Perpendicular to the direction of propagation
  - Magnitude is constant ("uniform") in any plane perpendicular to the propagation direction

$$\vec{J}_S = -J_x(t)\hat{a}_x$$

$$\vec{E} = E_x(z,t)\hat{a}_x$$

$$\vec{H} = H_y(z,t)\hat{a}_y$$



Our solution so far …

$$E_x(z,t) = Af\left(t - \frac{z}{v_p}\right) + Bg\left(t + \frac{z}{v_p}\right)$$

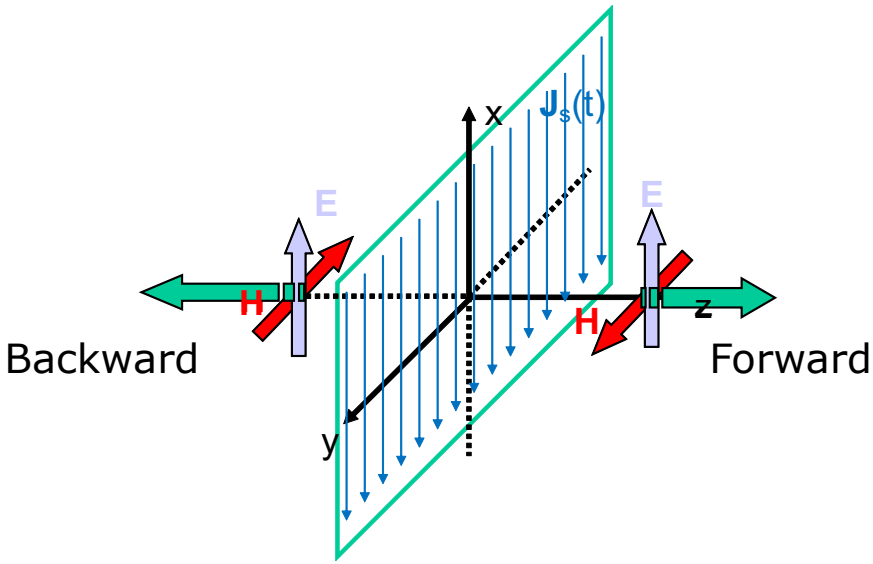
$$H_y(z,t) = \frac{1}{\eta_0}\left[Af\left(t - \frac{z}{v_p}\right) - Bg\left(t + \frac{z}{v_p}\right) \right]$$

Solution is a superposition of traveling waves

- one going in +z direction
- one going in -z direction

Plane Wave Propagation

- Generated waves travel in the +z direction for z>0 and –z direction for z<0



$z < 0$ then A=0

$$E_x(z,t) = Bg(t + \frac{z}{v_p})$$

$$H_y(z,t) = -\frac{B}{\eta_0} g(t + \frac{z}{v_p})$$

$z > 0$ then B=0

$$E_x(z,t) = Af(t - \frac{z}{v_p})$$

$$H_y(z,t) = \frac{A}{\eta_0} f(t - \frac{z}{v_p})$$

Next: Use boundary conditions to solve for A and B

Apply Boundary Condition

Boundary Condition

$$E_{1t} = E_{2t}$$

$\vec{E}_t$ is continuous

$$E_x(z,t) = Bg(t + \frac{z}{v_p}) \qquad z < 0$$

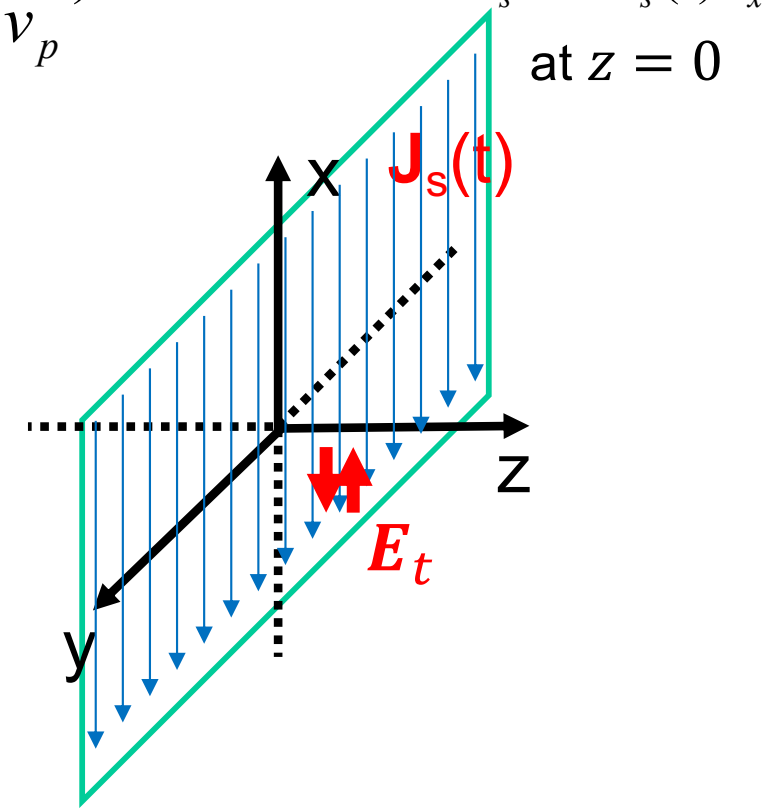
$$E_x(z,t) = Af(t - \frac{z}{v_p}) \qquad z > 0$$

$$\vec{J}_s = -J_s(t)\hat{a}_x \quad \text{at } z = 0$$

$$Af(t) = Bg(t)$$

$$E_{1t} = E_{2t}$$

at z=0



Eliminate Bg() in favor of Af()



propagation



propagation

$$z = 0^-$$

$$H_y(z,t) = \frac{A}{\eta_0} f(t)$$

$$H_y(z,t) = -\frac{A}{\eta_0} f(t)$$

$$z = 0^+$$

$$E_x(z,t) = Af(t)$$

$$H_y(z,t) = \frac{A}{\eta_0} f(t)$$

Use H_t discontinuity to relate the solutions to the current sheet

Boundary Condition

$$\vec{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

$\vec{H}_t$ is discontinuous at sheet

$$\vec{J}_s = -J_s(t)\hat{a}_x \quad \text{at } z = 0$$

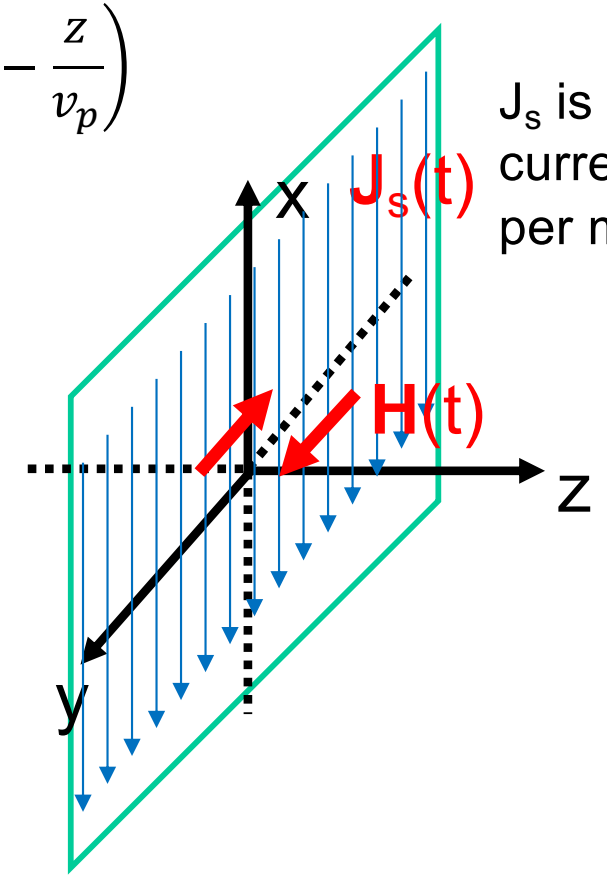
$$H_y(z,t) = -\frac{A}{\eta_0} f\left(t + \frac{z}{v_p}\right) \qquad z < 0$$

$$H_y(z,t) = \frac{A}{\eta_0} f\left(t - \frac{z}{v_p}\right) \qquad z > 0$$

$$\frac{A}{\eta_0} f(t) + \frac{A}{\eta_0} f(t) = J_s(t)$$

$$\frac{2A}{\eta_0} f(t) = J_s(t)$$

$$Af(t) = \frac{\eta_0}{2} J_s(t)$$



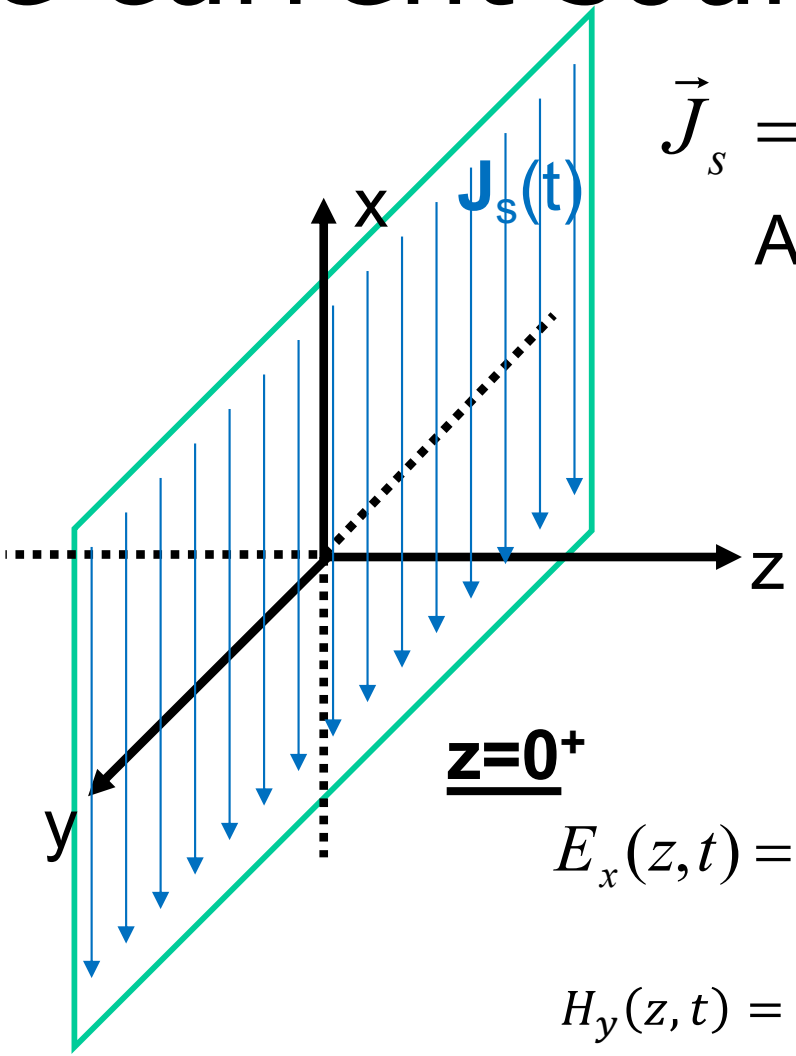
J_s is a surface current in Amps per meter

Final Step: Boundary conditions from the current source

We have to replace unknown functions f & g and constants A & B with something that relates them to the current source

$$\vec{J}_s = -J_s(t)\hat{a}_x$$

At z=0



z=0⁻

$$E_x(z,t) = Bg(t + \frac{z}{v_p})$$

$$H_y(z,t) = -\frac{A}{\eta_0} f\left(t + \frac{z}{v_p}\right)$$

$z < 0$

z=0⁺

$$E_x(z,t) = Af(t - \frac{z}{v_p})$$

$$H_y(z,t) = \frac{A}{\eta_0} f\left(t - \frac{z}{v_p}\right)$$

$z > 0$

The final answer at last

$z < 0$

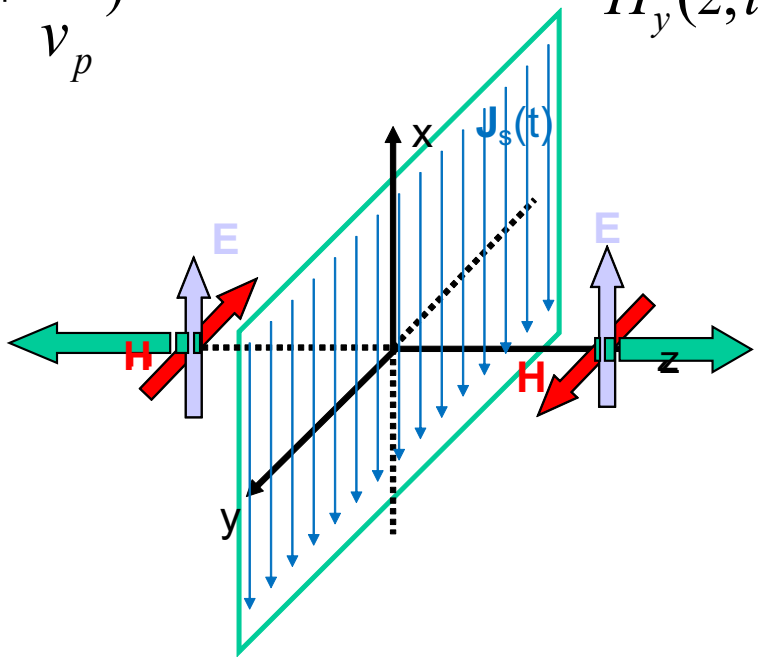
$$E_x(z,t) = \frac{\eta_0}{2} J_S(t + \frac{z}{v_p})$$

$$H_y(z,t) = -\frac{1}{2} J_S(t + \frac{z}{v_p})$$

$z > 0$

$$E_x(z,t) = \frac{\eta_0}{2} J_S(t - \frac{z}{v_p})$$

$$H_y(z,t) = \frac{1}{2} J_S(t - \frac{z}{v_p})$$

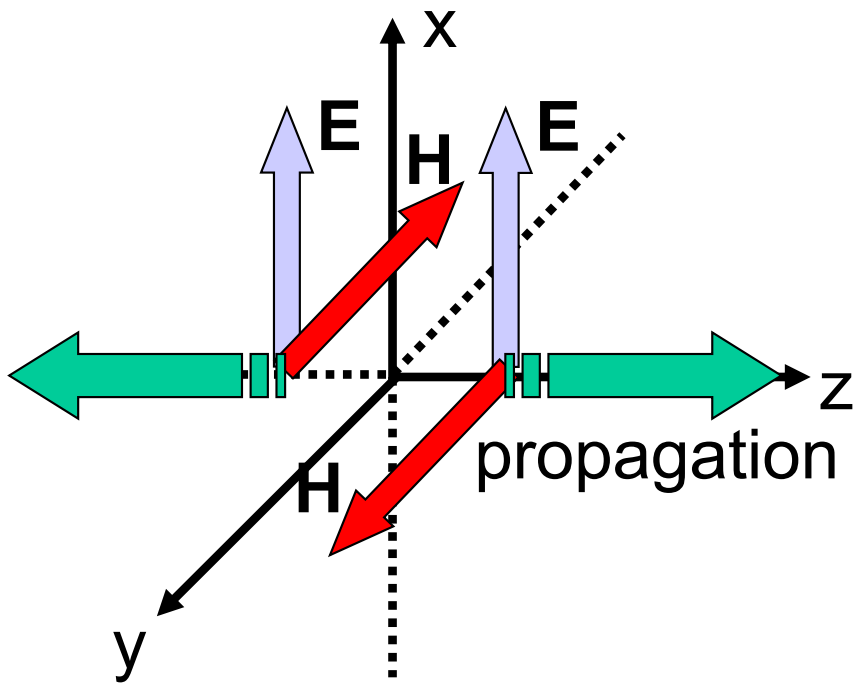


Combining the expressions…

$$\vec{E}(z,t) = \frac{\eta_0}{2} J_S(t \mp \frac{z}{v_p})\hat{a}_x$$

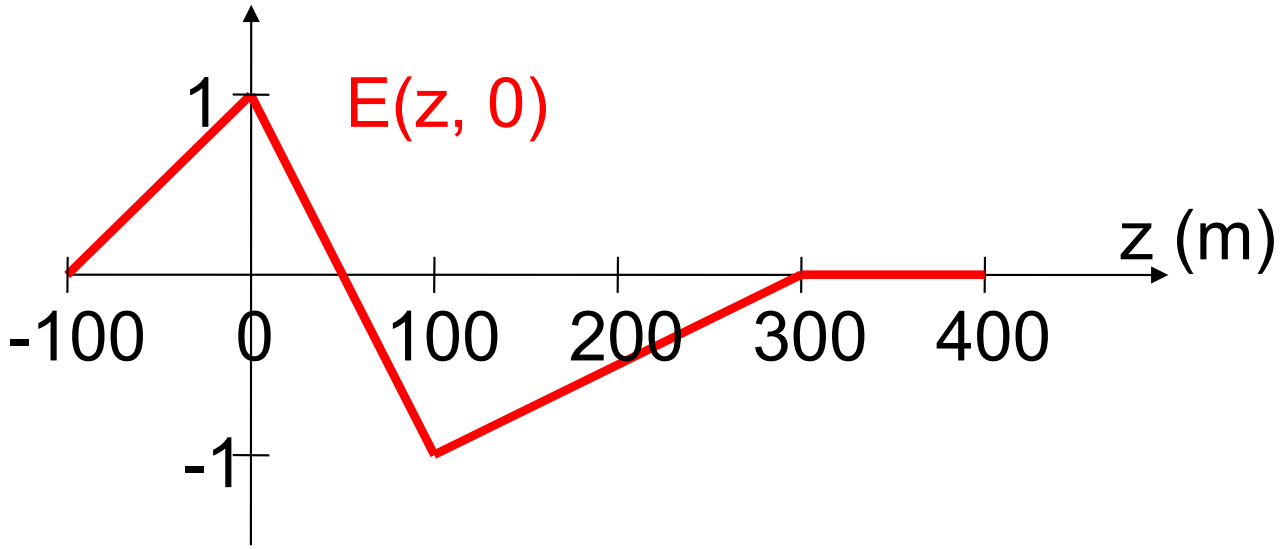
$$\vec{H}(z,t) = \pm\frac{1}{2} J_S(t \mp \frac{z}{v_p})\hat{a}_y$$

$z \gtrless 0$



Example: Transforming time and space

- A wave travelling in the $\mathbf{a}_z$ direction with speed 100m/s is measured at t=0:



Sketch:

(a) E(z, 1s)

(b) E(0, t)

(c) E(200m, t)

Lecture 19a Summary

- Wave emanates from z=0 so must travel $+a_z$ for z>0 and $-a_z$ for z<0

- Use integral form of Maxwell's Eqns to get BOUNDARY CONDITIONS

$$\vec{E}_t \text{ is continuous}$$

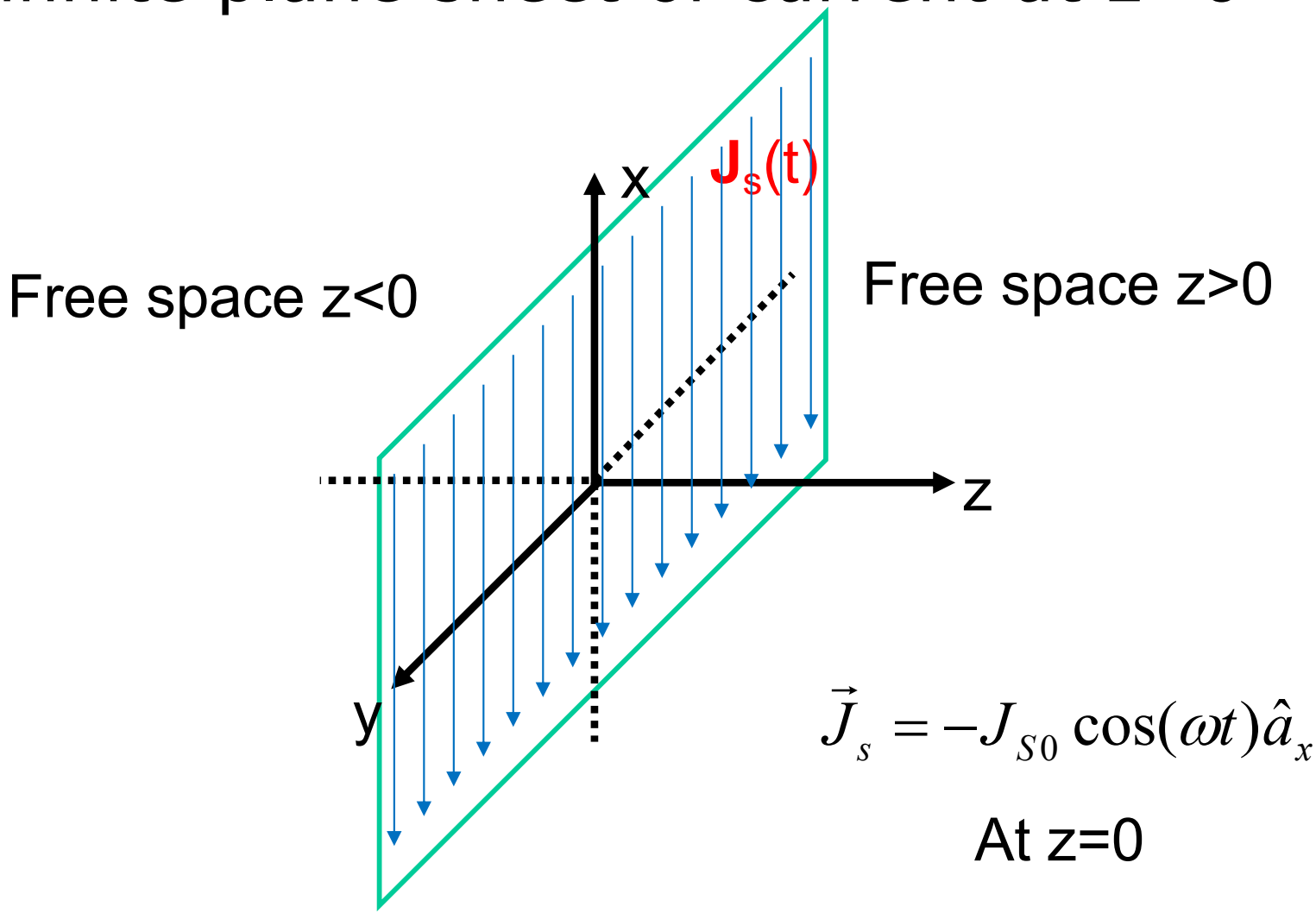
$$E_x(0^+) = E_x(0^-)$$

$$\vec{H}_t \text{ is discontinuous}$$

$$\mathrm{H_y}(0^+) - \mathrm{H_y}(0^-) = J_S$$

Sinusoidal Plane Waves

- Infinite plane sheet of current at z=0



$$\vec{J}_s = -J_{S0}\cos(\omega t)\hat{a}_x$$

At z=0

Time Domain Solution

$$\vec{J}_s = -J_{S0}\cos(\omega t)\hat{a}_x$$

$$\vec{E}(z,t) = \frac{\eta_0}{2} J_S(t \mp \frac{z}{v_p})\hat{a}_x$$

$$\vec{H}(z,t) = \pm\frac{1}{2} J_S(t \mp \frac{z}{v_p})\hat{a}_y$$

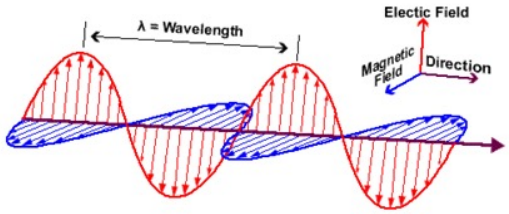
$$\vec{E}(z,t) = \frac{\eta_0 J_{S0}}{2}\cos(\omega t \mp \beta z)\hat{a}_x \qquad z \gtrless 0$$

$$\vec{H}(z,t) = \pm\frac{J_{S0}}{2}\cos(\omega t \mp \beta z)\hat{a}_y \qquad \beta = \frac{\omega}{v_p}$$

Phase constant

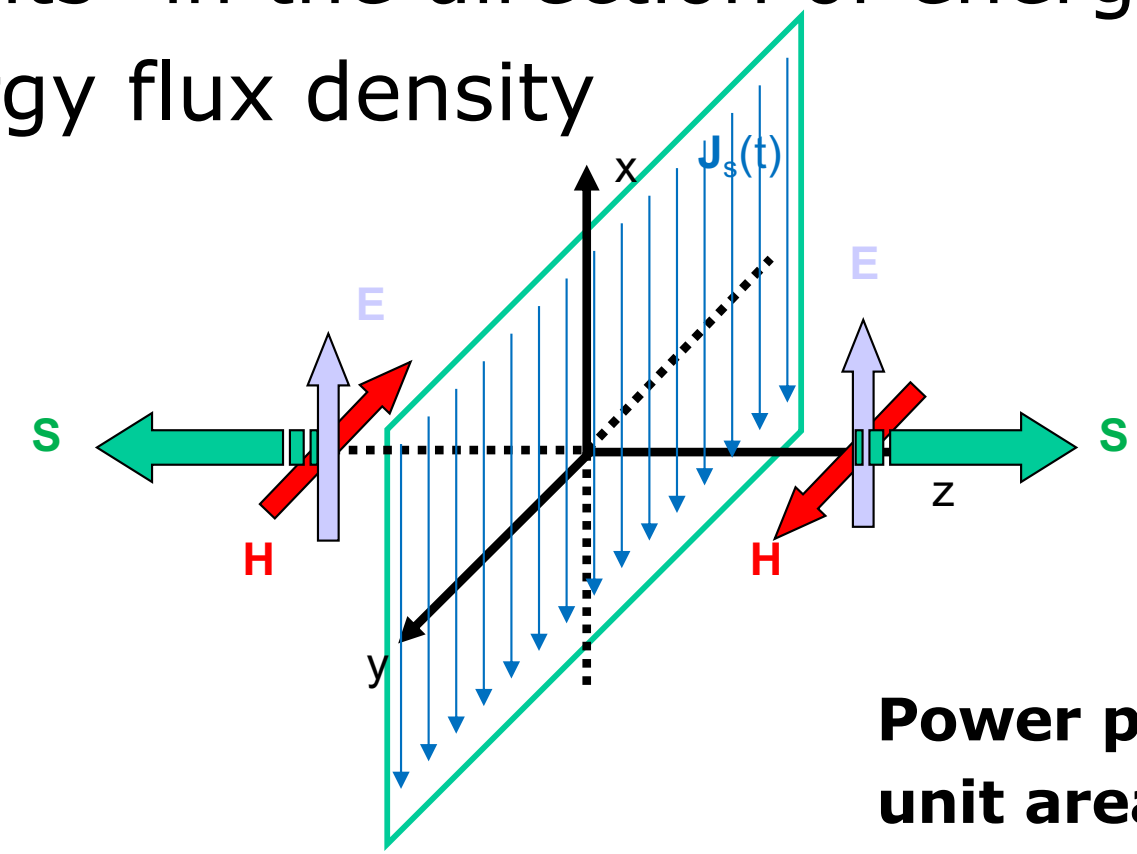
Wave Parameters

Electric Field	$\vec{E}(z,t) = \frac{\eta_0 J_{S0}}{2}\cos(\omega t \mp \beta z)\hat{a}_x$	(V/m)
Phase	$\phi = \omega t \mp \beta z$	(radians)
Angular Frequency	$\omega = \frac{\partial \phi}{\partial t}$	(radians/sec)
Linear Frequency	$f = \frac{\omega}{2\pi}$	(1/sec)
Phase Constant	$\beta = \left\lvert \frac{\partial \phi}{\partial z} \right\rvert$	(radians/m)
Wavelength	$\lambda = \frac{2\pi}{\beta}$	(m)
Phase Velocity	$v_p \equiv \frac{\omega}{\beta} = \lambda f = c$	(m/sec)
Impedance	$\eta_0 = \left\lvert \vec{E} \right\rvert / \left\lvert \vec{H} \right\rvert$	(Ω)



Poynting Vector

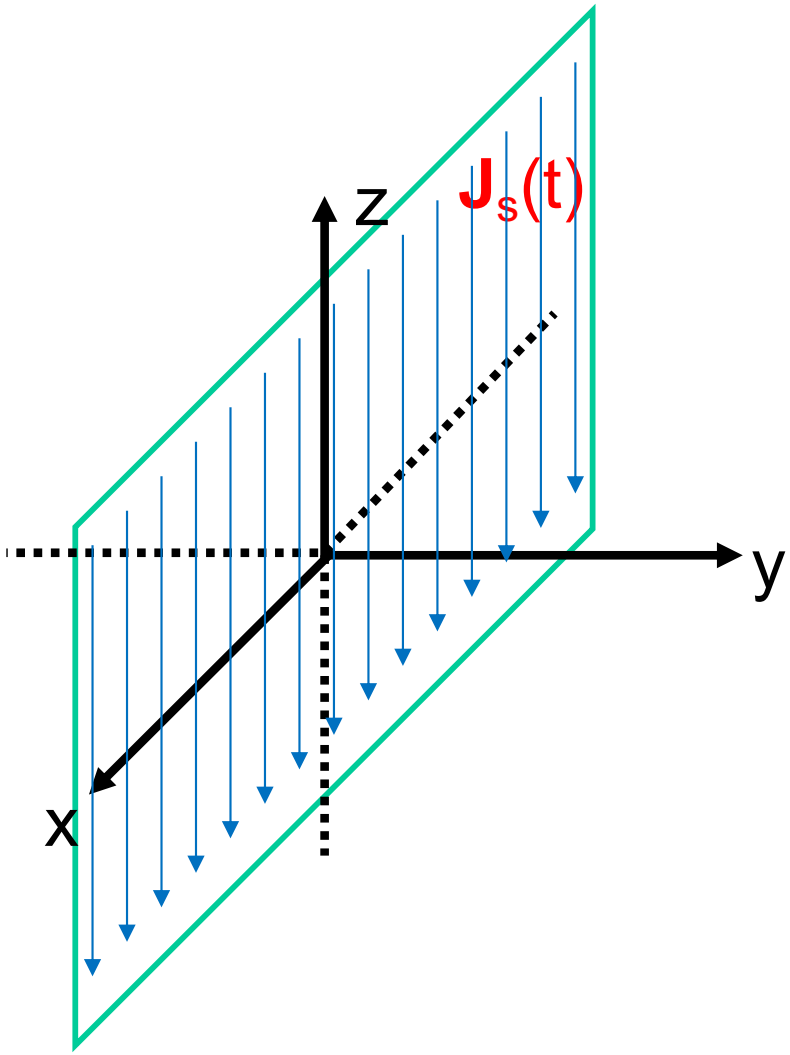
- "Points" in the direction of energy flow
- Energy flux density



Power per unit area

$$\vec{S} = \vec{E} \times \vec{H} = \pm \frac{\eta_0 J_{S0}^2}{4} \cos^2(\omega t \mp \beta z)\hat{a}_z \quad \text{W/m}^2 \quad z \gtrless 0$$

Example: Given $\vec{J}_s$ and $\vec{H}$, how to find $\vec{E}$?



$\vec{J}_s$ in –z direction, located on xz plane (y=0)

$\vec{H}$ is a sinusoidal function with angular frequency ω and magnitude H_0

What's the direction and magnitude of $\vec{E}$?

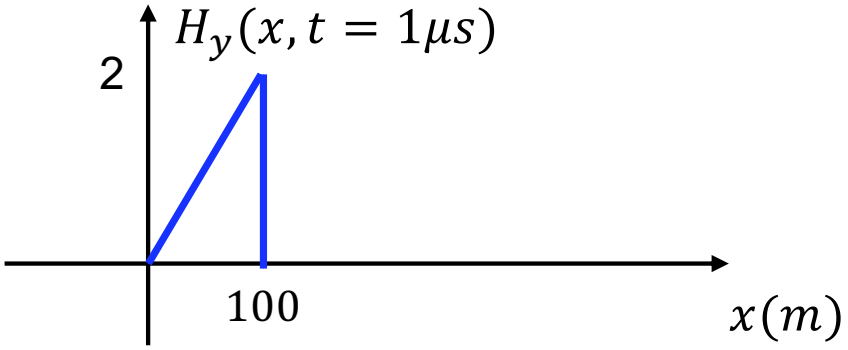
Hint: $\vec{E}\times\vec{H} \rightarrow \vec{k}$

$$\eta_0 = \frac{|\vec{E}|}{|\vec{H}|}$$

Example Problem: Wave propagation

Consider a time varying sheet current $\vec{J}_s$ in yz plane at $x = 0$. In $x > 0$ region, the magnetic field of the plane wave caused by the sheet current is $\vec{H}(x, t = 1\mu s) = 2x[u(x) - u(x - 100)]\hat{y}$ A/m.

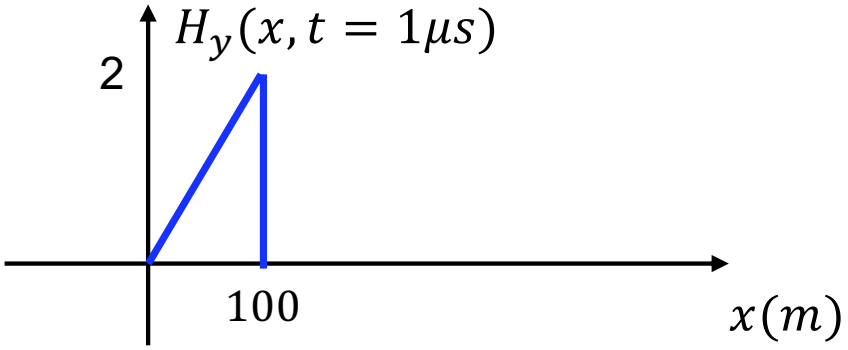
a) Plot $H_y(x = 400m, t)$ vs. t for $t \in [0, 10\ \mu s]$.



Example Problem: Wave propagation

Consider a time varying sheet current $\vec{J}_s$ in yz plane at $x = 0$. In $x > 0$ region, the magnetic field of the plane wave caused by the sheet current is $\vec{H}(x, t = 1\mu s) = 2x[u(x) - u(x - 100)]\hat{y}$ A/m.

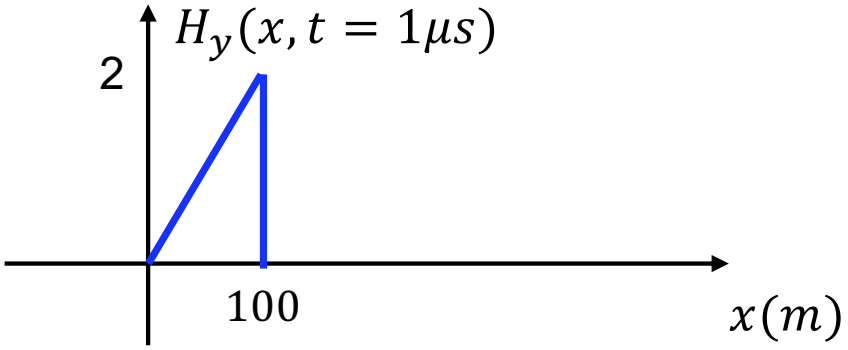
b) Plot $H_y(t = 4\mu s, x)$ vs. x for $x \in [-1500$ m, $+1500$ m$]$.



Example Problem: Wave propagation

Consider a time varying sheet current $\vec{J}_s$ in yz plane at $x = 0$. In $x > 0$ region, the magnetic field of the plane wave caused by the sheet current is $\vec{H}(x, t = 1\mu s) = 2x[u(x) - u(x - 100)]\hat{y}$ A/m.

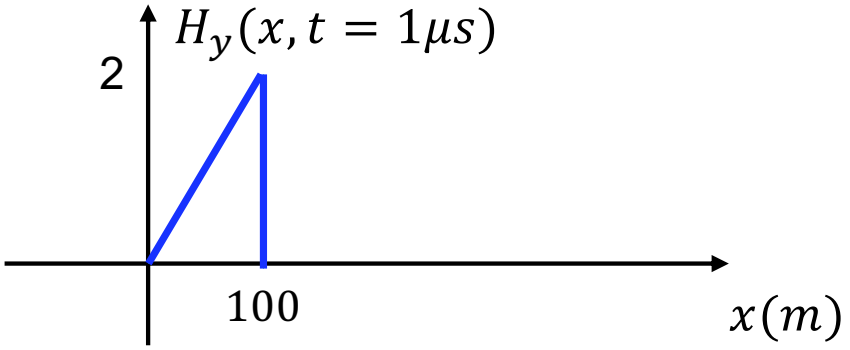
c) Plot $\vec{E}(t = 2\mu s, x)$ vs. x for $x \in$ [-500, 500m].



Example Problem: Wave propagation

Consider a time varying sheet current $\vec{J}_s$ in yz plane at $x = 0$. In $x > 0$ region, the magnetic field of the plane wave caused by the sheet current is $\vec{H}(x, t = 1\mu s) = 2x[u(x) - u(x - 100)]\hat{y}$ A/m.

d) Find the surface current density $\mathbf{J}_s(t)$ located at x = 0.



Lecture 19 Summary

- Wave propagation from current sheet

$$\vec{E}(z,t) = \frac{\eta_0}{2} J_S(t \mp \frac{z}{v_p})\hat{a}_x \qquad \vec{H}(z,t) = \pm\frac{1}{2} J_S(t \mp \frac{z}{v_p})\hat{a}_y$$

- Sinusoidal waves

$$\phi = \omega t \mp \beta z \qquad \omega = \frac{\partial \phi}{\partial t} \qquad f = \frac{\omega}{2\pi}$$

$$\beta = \mp\frac{\partial \phi}{\partial z} \qquad \lambda = \frac{2\pi}{\beta} \qquad v_p = \frac{\omega}{\beta}$$

$$v_p = \lambda f \qquad \left|\frac{E}{H}\right| = \eta_0$$